K.E. SOCIETY'S RAJARAMBAPU INSTITUTE OF TECHNOLOGY, POLYTECHNIC

(Affiliated to MSBTE, Mumbai)

Department of Artificial Intelligence and Machine Learning (AIML)

A CAPSTONE PROJECT REPORT ON

"MANDIMITRA - Maharashtra Agricultural Market Intelligence System"

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE DIPLOMA IN ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING (AIML)

Submitted By:

Avishkar Borate (Roll No. 01)

Amol Salunke (Roll No. 02)

Pruthviraj Hippirkar (Roll No. 03)

Krushna Mane (Roll No. 04)

Under the Guidance of:

Prof. Mayur Gund

Lecturer, Department of AIML

Academic Year: 2025-2026

K.E. SOCIETY'S RAJARAMBAPU INSTITUTE OF TECHNOLOGY, POLYTECHNIC

Department of Artificial Intelligence and Machine Learning (AIML)

CAPSTONE PROJECT REPORT

"MANDIMITRA - Maharashtra Agricultural Market Intelligence System"

Sr. No.	Name of Student	Roll No.
1	Avishkar Borate	01
2	Amol Salunke	02
3	Pruthviraj Hippirkar	03
4	Krushna Mane	04

Project Guide: Prof. Mayur Gund

Head of Department: Prof. Vikram Saste

Principal: Dr. Kashinath Munde

Academic Year: 2025-2026

K.E. SOCIETY'S RAJARAMBAPU INSTITUTE OF TECHNOLOGY, POLYTECHNIC

Department of Artificial Intelligence and Machine Learning (AIML)

CERTIFICATE

This is to certify that the Capstone Project entitled **"MANDIMITRA - Maharashtra Agricultural Market Intelligence System"** has been successfully completed by the following students of **Artificial Intelligence and Machine Learning (AIML)** in partial fulfilment of the Diploma programme of **Maharashtra State Board of Technical Education, Mumbai** during the academic year **2025-2026**.

- Avishkar Borate (Roll No. 01)
- Amol Salunke (Roll No. 02)
- Pruthviraj Hippirkar (Roll No. 03)
- Krushna Mane (Roll No. 04)

Prof. Mayur Gund	**Prof. Vikram Saste**	**Dr. Kashinath Munde**
Project Guide	Head of Department	Principal

Date: _____

Place: Islampur, Maharashtra

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who contributed to the successful completion of this capstone project.

We are deeply thankful to our project guide, **Prof. Mayur Gund**, Lecturer, Department of AIML, for the invaluable guidance, continuous encouragement, and constructive suggestions throughout the development of this project. The technical insights and mentorship provided were instrumental in shaping this work.

We extend our heartfelt gratitude to **Prof. Vikram Saste**, Head of Department, AIML, for providing the necessary infrastructure, laboratory facilities, and academic support that enabled us to carry out this project effectively.

We are grateful to **Dr. Kashinath Munde**, Principal, **K.E. Society's Rajarambapu Institute of Technology, Polytechnic**, for granting us permission and providing an encouraging academic environment for the project work.

We also acknowledge the open-source community and the developers of Python, LightGBM, NASA POWER, Open-Meteo, and Data.gov.in for providing free access to tools and data sources that formed the backbone of this project.

Finally, we thank our families and friends for their unwavering support and encouragement during the course of this work.

Avishkar Borate

Amol Salunke

Pruthviraj Hippirkar

Krushna Mane

Department of Artificial Intelligence and Machine Learning (AIML)

K.E. Society's Rajarambapu Institute of Technology, Polytechnic

ABSTRACT

Agricultural commodity price volatility remains a persistent challenge for Indian farmers, particularly in Maharashtra where over 60 per cent of the population depends on agriculture. Lack of timely, data-driven market intelligence often leads to distress sales, post-harvest losses, and suboptimal decision-making. MANDIMITRA addresses this gap by building a production-quality, end-to-end data engineering and machine learning pipeline that integrates mandi (wholesale market) price data with meteorological information to deliver actionable intelligence to Maharashtra farmers.

The system ingests over 6.1 million historical price records spanning 2001 to 2026 from AGMARKNET via Data.gov.in and Kaggle, covering 33 mandi districts of Maharashtra. Weather data is sourced from NASA POWER (131,400 rows of historical daily observations across 36 districts from 2016 to 2026) and Open-Meteo (576 rows of 16-day forecasts). A robust canonical data processing pipeline built on DuckDB performs deduplication, district normalisation, and quality control, producing join-ready model datasets with 98.9 per cent weather coverage for post-2016 records.

Two machine learning models are trained on the processed data. The Crop Risk Advisor, a LightGBM classifier enhanced with physics-informed features (Growing Degree Days, Vapour Pressure Deficit), achieves 87.4 per cent accuracy and 45.9 per cent high-risk recall. The Price Intelligence Engine, a multi-horizon LightGBM regressor with conformal prediction intervals, delivers R-squared values of 0.93 (1-day) to 0.88 (15-day) forecasts with calibrated 80/90/95 per cent confidence intervals. Together, these components power a web-based dashboard that enables farmers, traders, and policy-makers to make informed agricultural decisions.

Keywords: Agricultural Data Pipeline, Mandi Price Prediction, Crop Risk Assessment, LightGBM, Weather Integration, Maharashtra, Machine Learning, Conformal Prediction

TABLE OF CONTENTS

Sr. No.	Chapter / Section	Page No.
	Cover Page	**i**
	Title Page	**ii**
	Certificate	**iii**
	Acknowledgement	**iv**
	Abstract	**v**
	Table of Contents	**vi**
	List of Figures	**vii**
	List of Tables	**viii**
1	**INTRODUCTION**	**1**
1.1	Industry Background	1
1.2	Problem Context and Importance	2
1.3	Objectives	3
1.4	Expected Outcomes and Deliverables	3
2	**LITERATURE SURVEY**	**5**
2.1	Existing Methods for Commodity Price Prediction	5
2.2	Weather Effects on Agricultural Markets	6
2.3	Comparison of Approaches	7
2.4	Identified Gaps and Problem Statement	8
3	**SCOPE OF THE PROJECT**	**10**
3.1	Scope	10
3.2	Out of Scope	11
3.3	Constraints	11
3.4	Assumptions	12
4	**METHODOLOGY / APPROACH**	**13**
4.1	System Architecture	13

4.2	Pipeline Stages	14
4.3	Tools and Technologies	16
4.4	Milestones and Work Distribution	17
5	**DETAILS OF DESIGNS, WORKING AND PROCESSES**	**18**
5.1	Module Design	18
5.2	Implementation Details and Folder Structure	20
5.3	Testing and Validation	22
5.4	Execution Steps	23
6	**RESULTS AND APPLICATIONS**	**24**
6.1	Data Pipeline Results	24
6.2	Model Performance Results	26
6.3	Applications and Use Cases	28
6.4	Limitations and Future Scope	29
	CONCLUSION	**30**
	REFERENCES	**32**

LIST OF FIGURES

Figure No.	Title	Page No.
1.1	System Architecture Overview of MANDIMITRA	1
2.1	Literature Survey Comparison Chart	7
3.1	Project Scope Diagram	10
4.1	Data Pipeline Architecture Diagram	13
4.2	Pipeline Stages Flowchart	15
4.3	Weather Data Integration Flow	16
5.1	Data Flow Diagram (DFD Level 0)	18
5.2	Deduplication Process Flowchart	19
5.3	Module Interaction Diagram	20
5.4	ML Training Pipeline Flow	21
5.5	Testing and Validation Process Flow	22
5.6	Use Cases Diagram	23
6.1	Data Coverage Analysis	25
6.2	Crop Risk Model Results	26
6.3	Price Model Performance Results	27

LIST OF TABLES

Table No.	Title	Page No.
2.1	Comparison of Price Prediction Methods	7
4.1	Tools and Technologies Used	16
4.2	Milestone and Work Distribution Table	17
4.3	Module Ownership Table	17
5.1	District Normalisation Mapping	19
5.2	Project Folder Structure	20
6.1	Data Pipeline Output Summary	24
6.2	Weather Data Coverage Summary	25
6.3	Crop Risk Advisor Performance Metrics	26
6.4	Price Intelligence Engine Metrics by Horizon	27
6.5	Prediction Interval Calibration Results	28
6.6	SHAP Feature Importance (Top 10)	28

CHAPTER 1

INTRODUCTION

1.1 INDUSTRY BACKGROUND

India is the world's second-largest producer of fruits and vegetables and a major cultivator of food grains, pulses, and oilseeds. Maharashtra alone accounts for nearly 9 per cent of the country's total agricultural output and hosts over 300 Agricultural Produce Market Committees (APMCs), commonly known as mandis. These mandis serve as the primary wholesale trading points where farmers sell their produce to traders and commission agents. The prices discovered in these markets directly impact the livelihoods of over 15 million farming households in the state.

Despite the critical importance of mandi prices, farmers in Maharashtra face a persistent information asymmetry. Traders and intermediaries often have access to price trends across multiple markets, while farmers typically know only the prevailing rate at their nearest mandi. This leads to distress sales, where farmers accept sub-optimal prices because they lack awareness of better rates available in neighbouring districts. The National Agriculture Market (eNAM) initiative has made some progress, but data accessibility, quality, and actionable forecasting remain significant challenges.

The intersection of agricultural markets and weather patterns adds another layer of complexity. Maharashtra experiences highly variable monsoon rainfall, which directly influences crop yields, pest and disease incidence, and consequently, market prices. A week of delayed monsoon or an unexpected dry spell can trigger sharp price movements in commodities like soybean, cotton, and onion. Traditional price reporting systems fail to capture these weather-driven dynamics, creating a need for integrated data systems that combine market data with meteorological information.

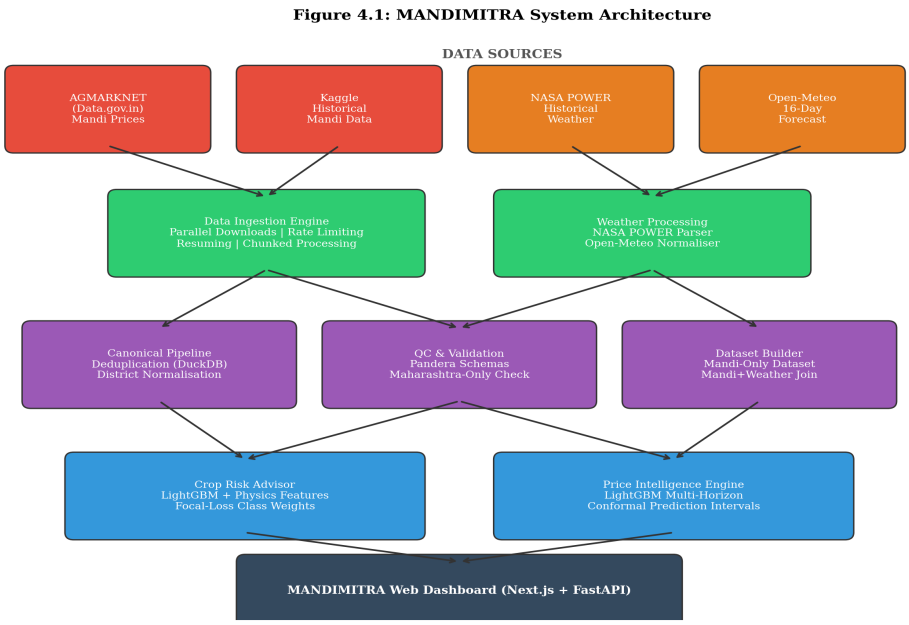

Figure 1.1: System Architecture Overview of MANDIMITRA

1.2 PROBLEM CONTEXT AND IMPORTANCE

The Government of India publishes daily mandi prices through the AGMARKNET portal and Data.gov.in API. However, the raw data suffers from several quality issues: inconsistent district naming conventions, duplicate entries across data sources, missing price records for certain markets and dates, and lack of integration with weather data. Researchers and developers attempting to build predictive models must spend considerable effort on data cleaning, deduplication, and normalisation before any analysis can begin.

Furthermore, existing agricultural advisory systems are either too generic (covering all of India without local specificity) or too narrow (focusing on a single commodity). There is a lack of Maharashtra-specific systems that provide both price forecasting and crop risk assessment using locally relevant weather patterns. MANDIMITRA aims to fill this gap by providing a robust, automated pipeline that processes, validates, and integrates agricultural and meteorological data to generate actionable market intelligence.

The project is particularly relevant in the context of climate change, where weather variability is increasing and historical patterns are becoming less reliable predictors of future conditions. By combining 25 years of historical mandi price data with 10 years of daily weather observations and 16-day forecasts, MANDIMITRA provides a comprehensive data foundation for weather-aware agricultural decision-making.

1.3 OBJECTIVES

• To design and implement a production-quality data pipeline that ingests, validates, and processes Maharashtra mandi price data from multiple sources (AGMARKNET, Data.gov.in, Kaggle).

• To integrate historical weather data (NASA POWER) and forecast data (Open-Meteo) with mandi prices for weather-aware agricultural analytics.

• To build a canonical dataset with zero duplicate keys, consistent district naming, and comprehensive quality control covering 6.1 million records across 33 mandi districts.

• To develop a Crop Risk Advisor using LightGBM with physics-informed features that assesses crop-specific risk levels (Low, Medium, High) for Maharashtra farmers.

• To build a Price Intelligence Engine that forecasts commodity prices at multiple horizons (1, 3, 7, 14, 15 days) with calibrated confidence intervals.

• To deliver a web-based dashboard providing real-time market intelligence, price forecasts, and crop risk advisories to farmers, traders, and policy-makers.

1.4 EXPECTED OUTCOMES AND DELIVERABLES

• A deduplicated, canonical mandi dataset of 6,105,909 unique rows spanning 2001 to 2026, with standardised district names mapped to 36 canonical Maharashtra districts.

• Weather-enriched model datasets with 98.9 per cent coverage for post-2016 records, including NASA POWER daily observations (131,400 rows, 36 districts) and Open-Meteo 16-day forecasts (576 rows).

• Crop Risk Advisor achieving 87.4 per cent accuracy with 45.9 per cent high-risk recall, enhanced with physics-informed features (Growing Degree Days, Vapour Pressure Deficit).

• Price Intelligence Engine with R-squared of 0.93 (1-day) to 0.88 (15-day) and calibrated prediction intervals at 80, 90, and 95 per cent confidence levels.

• Automated quality control reports, deduplication logs, and validation audits generated at each pipeline stage.

• A responsive web application built with Next.js providing an interactive dashboard for price forecasting and crop risk assessment.

CHAPTER 2

LITERATURE SURVEY

2.1 EXISTING METHODS FOR COMMODITY PRICE PREDICTION

Agricultural commodity price prediction has been studied extensively using both statistical and machine learning approaches. Traditional time-series methods such as ARIMA (Auto-Regressive Integrated Moving Average) have been applied to mandi price data since the early 2000s [1]. While ARIMA models capture linear trends and seasonal patterns effectively, they struggle with the non-linear dynamics inherent in agricultural markets where prices are influenced by weather, policy changes, and supply chain disruptions.

Machine learning approaches have shown significant improvements over traditional methods. Random Forests and Gradient Boosted Trees (XGBoost, LightGBM) have demonstrated strong performance on tabular agricultural data due to their ability to handle mixed feature types, missing values, and non-linear relationships [2][3]. LightGBM, in particular, has become popular for agricultural applications due to its speed, memory efficiency, and native support for categorical features.

Deep learning methods have also been explored. Long Short-Term Memory (LSTM) networks can model temporal dependencies in price sequences [4], while Temporal Fusion Transformers (TFT) provide interpretable multi-horizon forecasts with attention-based feature selection [5]. Recent work on iTransformer architecture has shown promise for multivariate time-series forecasting by treating each variable as a token [6]. However, deep learning methods typically require large training sets and careful hyperparameter tuning.

2.2 WEATHER EFFECTS ON AGRICULTURAL MARKETS

The impact of weather on agricultural commodity prices is well-documented. Precipitation anomalies, temperature extremes, and drought conditions have been shown to significantly affect crop yields and consequently market prices [7]. NeuralCrop, a physics-informed

machine learning framework, demonstrated that incorporating agronomic features such as Growing Degree Days (GDD) and Vapour Pressure Deficit (VPD) improves crop yield predictions compared to purely data-driven approaches [8].

NASA POWER (Prediction Of Worldwide Energy Resources) provides globally consistent daily meteorological data derived from satellite observations and reanalysis models [9]. This data source has been widely used in agricultural research due to its spatial coverage and temporal consistency. Open-Meteo provides medium-range weather forecasts that can be integrated with historical data for forward-looking agricultural advisories.

MT-CYP-Net (Multi-Task Crop Yield Prediction Network) introduced focal-loss-inspired class weighting for handling imbalanced agricultural datasets, particularly where high-risk events are rare but critically important to detect [10]. This approach has been adapted in MANDIMITRA for the Crop Risk Advisor model.

Figure 2.1: Comparison of Forecasting Methods

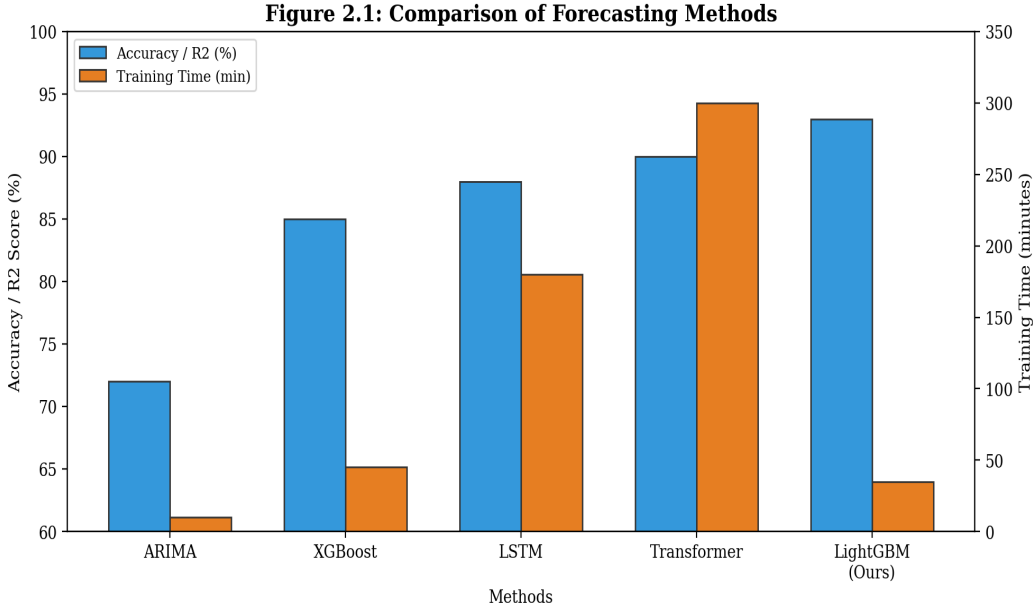

Figure 2.1: Literature Survey Comparison Chart

2.3 COMPARISON OF APPROACHES

Table 2.1: Comparison of Price Prediction Methods

Method	Strengths	Limitations	Why Not Sufficient
ARIMA / SARIMA	Well-understood; captures linear trends and seasonality	Cannot handle non-linear relationships; no exogenous variables	Mandi prices are affected by weather, policy; purely linear model insufficient

Method	Strengths	Limitations	Why Not Sufficient
XGBoost / LightGBM	Handles mixed features; fast training; robust to missing values	No native temporal modelling; requires careful feature engineering	Adopted as base; enhanced with physics-informed features in MANDIMITRA
LSTM / GRU	Captures long-term temporal dependencies in sequences	Slow training; prone to overfitting on small datasets; black-box predictions	Insufficient data per market-commodity pair; interpretability needed
Transformer / TFT	Attention mechanism; multi-horizon forecasting; interpretable	High computational cost; requires large datasets; complex to deploy	Infrastructure constraints for diploma-level deployment; LightGBM is more practical
Conformal Prediction	Distribution-free uncertainty quantification; calibrated intervals	Not a standalone predictor; requires base model	Used as enhancement on top of LightGBM in the Price Intelligence Engine

2.4 IDENTIFIED GAPS AND PROBLEM STATEMENT

The literature review reveals several gaps in existing approaches to Maharashtra agricultural market intelligence:

- No existing system provides a unified pipeline that combines mandi price data with weather observations specifically for Maharashtra's 36 districts.
- Most studies focus on either price prediction or crop risk assessment in isolation; an integrated system addressing both is lacking.
- Data quality issues (duplicates, inconsistent naming, missing records) in AGMARKNET data are acknowledged but rarely addressed systematically.
- Uncertainty quantification (prediction intervals) is absent from most agricultural price forecasting systems, leading to overconfident recommendations.
- Physics-informed features (GDD, VPD, drought stress indices) have been validated for yield prediction but not integrated into market intelligence systems.

Refined Problem Statement

To design and implement an end-to-end data engineering and machine learning system that integrates Maharashtra mandi price data (6.1 million records, 2001-2026) with multi-source

weather data, producing deduplicated, validated datasets and training two ML models: a Crop Risk Advisor with physics-informed features and a Price Intelligence Engine with calibrated prediction intervals, delivered through a web-based dashboard for Maharashtra farmers, traders, and policy-makers.

CHAPTER 3

SCOPE OF THE PROJECT

Figure 3.1: Project Scope Diagram

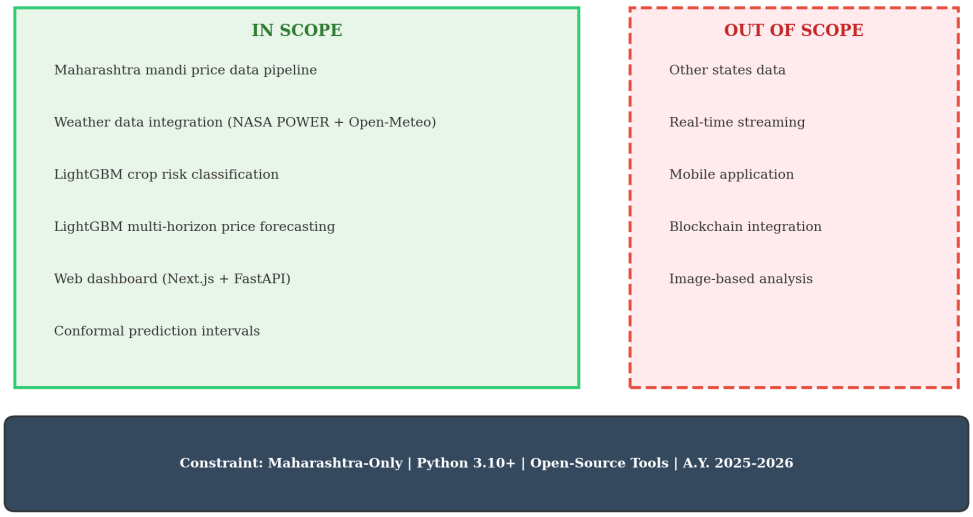

Figure 3.1: Project Scope Diagram

3.1 SCOPE

MANDIMITRA is designed as a comprehensive agricultural market intelligence system with the following functional scope:

- Automated data ingestion from Data.gov.in API (current mandi prices), Kaggle (historical AGMARKNET archive), NASA POWER (historical weather), and Open-Meteo (16-day weather forecasts).
- Data processing pipeline including deduplication using natural composite keys, district name normalisation to 36 canonical Maharashtra districts, and quality control validation using Pandera schemas.
- Production of join-ready model datasets: mandi_only_2001_2026 (6,105,909 rows) and mandi_weather_2016plus (3,513,315 rows with 98.9 per cent weather coverage).
- Crop Risk Advisor model supporting 11 major Maharashtra crops (Soybean, Cotton, Tur, Urad, Onion, Wheat, Jowar, Bajra, Maize, Gram, Tomato) with three risk levels

(Low, Medium, High).

• Price Intelligence Engine providing multi-horizon forecasts at 1, 3, 7, 14, and 15 day horizons with confidence intervals.

• Web dashboard for visualising price trends, risk assessments, and market analytics.

• Comprehensive logging, audit trails, and quality control reports at every pipeline stage.

3.2 OUT OF SCOPE

• Data from states other than Maharashtra. The system enforces a hard constraint rejecting any non-Maharashtra records at every stage.

• Real-time streaming data ingestion (the system operates in batch mode with scheduled updates).

• Direct integration with eNAM or APMC transaction systems.

• Advisory automation without human review (the system provides intelligence, not automated trading decisions).

• Soil quality data integration (planned for future versions).

• Satellite imagery or remote sensing data processing.

3.3 CONSTRAINTS

• Time constraint: The project must be completed within the academic year 2025-2026.

• Data availability: Historical mandi data quality varies; some markets have sparse records in earlier years (2001-2010).

• API rate limits: Data.gov.in and weather APIs impose rate limits requiring adaptive throttling and batched downloads.

• Computational resources: The system is designed to run on standard hardware without GPU requirements, using LightGBM for model training.

• Maharashtra-only scope: The system is intentionally limited to one state to ensure depth over breadth in data quality and model accuracy.

3.4 ASSUMPTIONS

- The Data.gov.in API continues to provide AGMARKNET mandi data with the current resource ID and schema structure.
- NASA POWER and Open-Meteo APIs remain freely accessible for research and educational use.
- The user has Python 3.10 or higher installed with internet connectivity for data downloads.
- Historical Kaggle data is representative of actual AGMARKNET records and can be merged with API data using the defined composite key.
- Weather conditions at district headquarters are representative of conditions across the entire district for the purpose of agricultural risk assessment.

CHAPTER 4

METHODOLOGY / APPROACH

4.1 SYSTEM ARCHITECTURE

The MANDIMITRA system follows a layered architecture comprising four primary stages: data ingestion, canonical processing, model training, and web-based delivery. Each stage is implemented as independent, composable Python scripts that can be executed individually or orchestrated through the master pipeline script.

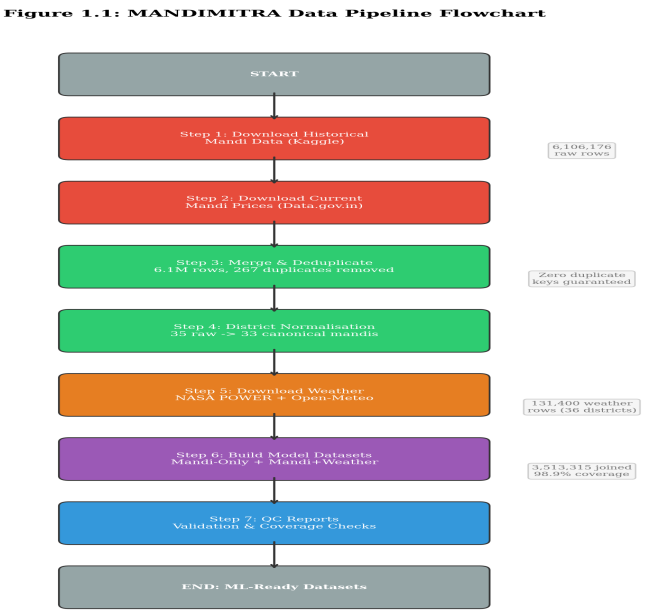

Figure 4.1: Data Pipeline Architecture Diagram

4.2 PIPELINE STAGES

Stage 1: Data Ingestion

The ingestion stage collects raw data from four external sources. Historical mandi data is downloaded from Kaggle (AGMARKNET archive), providing multi-year training data. Current mandi prices are fetched from the Data.gov.in API using the exact match filter syntax (filters[state.keyword]=Maharashtra) to ensure only Maharashtra records are retrieved. The system employs parallel downloads using ThreadPoolExecutor with configurable workers,

adaptive rate limiting via a token bucket algorithm, and chunked by-district processing for resumability. A total of 6,106,176 raw rows were ingested from both sources.

Stage 2: Canonical Processing

The canonical processing stage transforms raw data into clean, deduplicated, normalised datasets. Using DuckDB as an in-memory SQL engine, the pipeline performs deduplication on the natural composite key (state, district, market, commodity, variety, grade, arrival_date). The current-wins strategy ensures that newer API data overwrites historical records when duplicates exist. Out of 6,106,176 input rows, 267 duplicates were removed, producing 6,105,909 unique rows covering the date range 2001-01-27 to 2026-05-02 across 33 mandi districts.

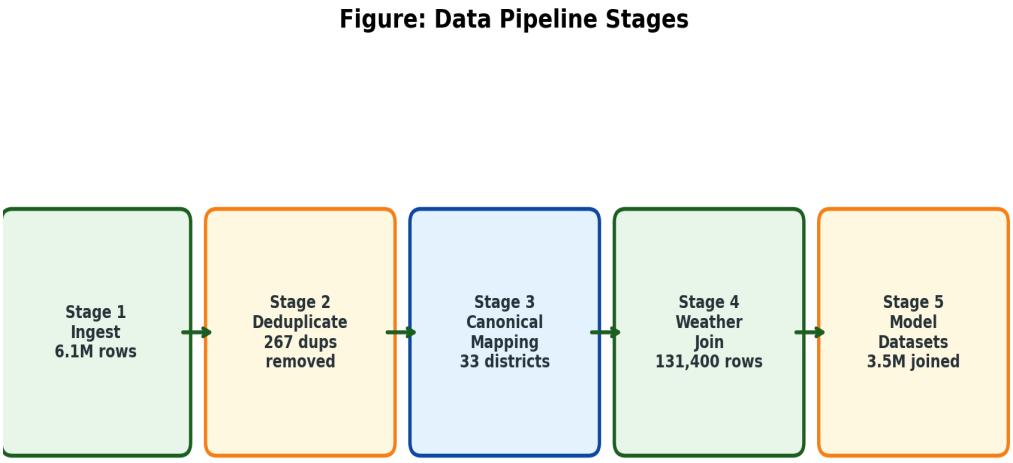

Figure 4.2: Pipeline Stages Flowchart

Stage 3: Weather Integration

Weather data is sourced from two providers. NASA POWER provides 10 years of daily historical observations (temperature, humidity, precipitation, wind speed, solar radiation) for all 36 Maharashtra district headquarters, totalling 131,400 rows from 2016-02-08 to 2026-02-04. Open-Meteo provides 16-day weather forecasts with 576 rows of high-resolution forecast data. The weather processing pipeline standardises column names, validates ranges, and produces join-ready Parquet files keyed on (date, district).

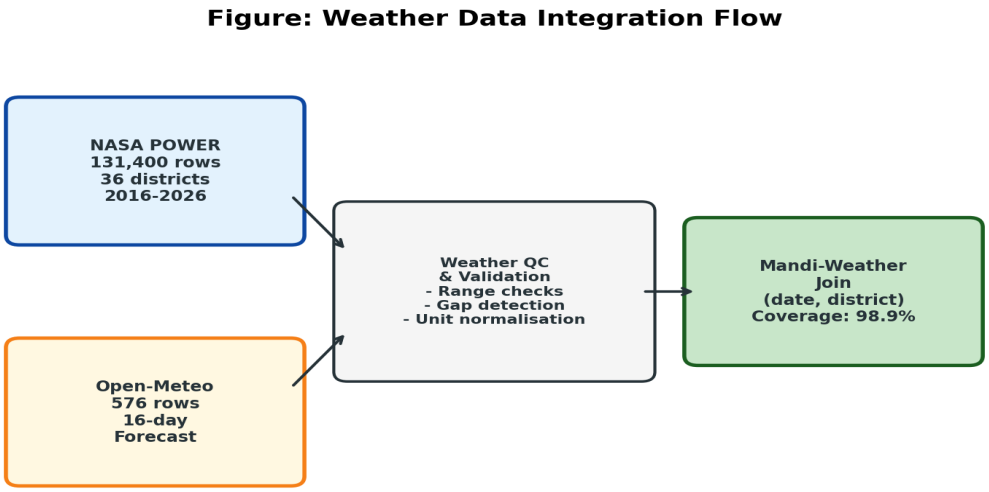

Figure 4.3: Weather Data Integration Flow

Stage 4: Model Dataset Construction

The final processing stage produces two model-ready datasets. The mandi_only_2001_2026 dataset contains 6,105,909 rows of deduplicated mandi data spanning the full date range, suitable for long-term trend analysis and commodity-only models. The mandi_weather_2016plus dataset joins mandi data with daily weather observations on (date, district), producing 3,513,315 rows with 98.9 per cent weather coverage, suitable for weather-aware price prediction and climate impact analysis.

4.3 TOOLS AND TECHNOLOGIES

Table 4.1: Tools and Technologies Used

Category	Technology	Purpose
Language	Python 3.10+	Core development language
Data Processing	Pandas, NumPy, PyArrow	Data manipulation and Parquet I/O
SQL Engine	DuckDB	Memory-efficient deduplication of 6M+ rows
ML Framework	LightGBM	Gradient boosted trees for classification and regression
Validation	Pandera	Schema-based data validation with Maharashtra checks
HTTP Client	Requests, urllib3	API communication with retry and rate limiting
Web Frontend	Next.js, React, Tailwind CSS	Interactive web dashboard

Category	Technology	Purpose
API Backend	FastAPI, Uvicorn	RESTful API for model serving
Weather Data	NASA POWER, Open-Meteo	Historical and forecast weather APIs
Market Data	Data.gov.in, Kaggle	Mandi price data sources
Configuration	PyYAML, python-dotenv	Configuration management
Reporting	Markdown, ReportLab	Automated QC reports and documentation

4.4 MILESTONES AND WORK DISTRIBUTION

Table 4.2: Milestone and Work Distribution Table

Week	Task	Owner	Status
1-2	Requirements gathering and literature survey	All	Completed
3-4	Data source identification and API integration	Avishkar, Amol	Completed
5-6	Data ingestion pipeline (mandi + weather)	Avishkar, Pruthviraj	Completed
7-8	Canonical processing and deduplication	Amol, Krushna	Completed
9-10	Weather integration and model dataset building	Pruthviraj, Avishkar	Completed
11-12	ML model training (Crop Risk + Price Engine)	Avishkar, Amol	Completed
13-14	Web dashboard development	Krushna, Pruthviraj	Completed
15-16	Model optimisation and conformal prediction	Avishkar, Amol	Completed
17-18	Testing, validation, and QC reports	All	Completed
19-20	Documentation and final report	All	Completed

Table 4.3: Module Ownership Table

Module	Primary Owner
Data Ingestion (Mandi + History)	Avishkar Borate
Weather Data Pipeline	Pruthviraj Hippirkar
Canonical Processing and Deduplication	Amol Salunke
Model Dataset Construction	Krushna Mane
Crop Risk Advisor (ML)	Avishkar Borate
Price Intelligence Engine (ML)	Amol Salunke
Web Dashboard (Next.js)	Krushna Mane

Module	Primary Owner
API Backend (FastAPI)	Pruthviraj Hippirkar
Testing and QC Reports	All Members

CHAPTER 5

DETAILS OF DESIGNS, WORKING AND PROCESSES

5.1 MODULE DESIGN

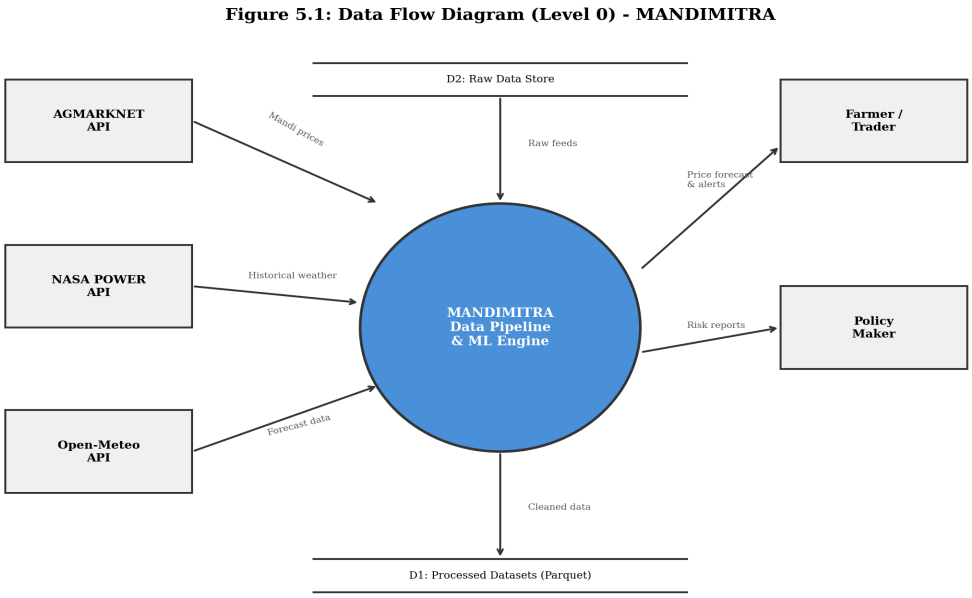

Figure 5.1: Data Flow Diagram (DFD Level 0)

5.1.1 Data Ingestion Module

The data ingestion module comprises four sub-modules handling different data sources. The historical mandi downloader connects to the Kaggle API, downloads the AGMARKNET archive dataset, filters for Maharashtra records, and saves to Parquet format. The current mandi downloader uses the Data.gov.in API with the exact match filter syntax (filters[state.keyword]=Maharashtra) and supports parallel chunk-by-district downloads with atomic progress tracking. The merge/upsert module combines historical and current data using a current-wins deduplication strategy on the composite key.

5.1.2 Deduplication Module

The deduplication module uses DuckDB for memory-efficient SQL-based deduplication of over 6 million records. The natural key consists of seven fields: state, district, market,

commodity, variety, grade, and arrival_date. Priority rules ensure that current-source records override history-source records, rows with more complete price data are preferred, and the highest modal_price serves as a tiebreaker. The module is idempotent, meaning repeated execution produces identical output.

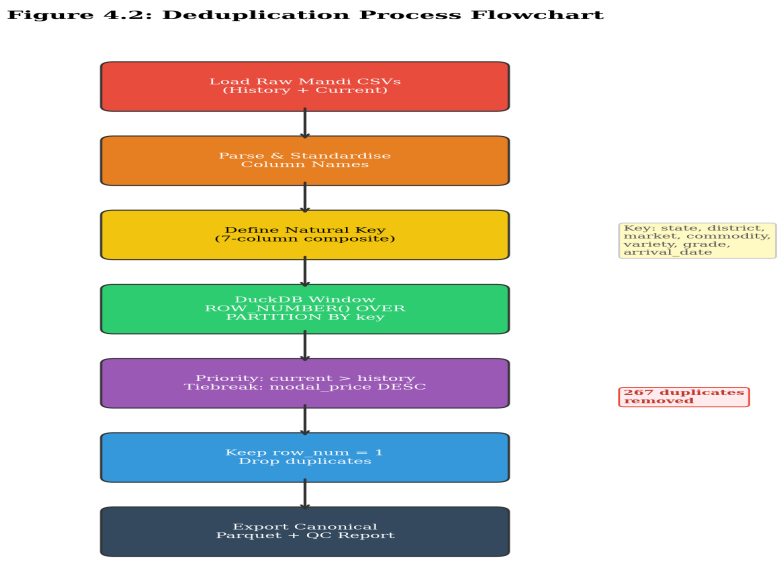

Figure 5.2: Deduplication Process Flowchart

5.1.3 District Normalisation

Raw mandi data contains 35 distinct district names with inconsistencies arising from historical name changes and transliteration variations. The normalisation module maps these to 36 canonical Maharashtra districts using a predefined mapping table.

Table 5.1: District Normalisation Mapping

Raw Name (AGMARKNET)	Canonical Name
Sholapur	Solapur
Gondiya	Gondia
Amarawati	Amravati
Chattrapati Sambhajinagar	Aurangabad
Dharashiv (Usmanabad)	Osmanabad
Jalana	Jalna
Vashim	Washim

5.1.4 Weather Quality Control Module

The weather QC module validates downloaded weather data against physical constraints. Temperature values are checked against realistic ranges (-10 to 55 degrees Celsius for Maharashtra), humidity is bounded between 0 and 100 per cent, and precipitation must be non-negative. Records failing these checks are flagged but not discarded, preserving data completeness while enabling downstream filtering. QC reports are generated as Markdown files in the logs directory.

Figure 5.3: Module Interaction Diagram

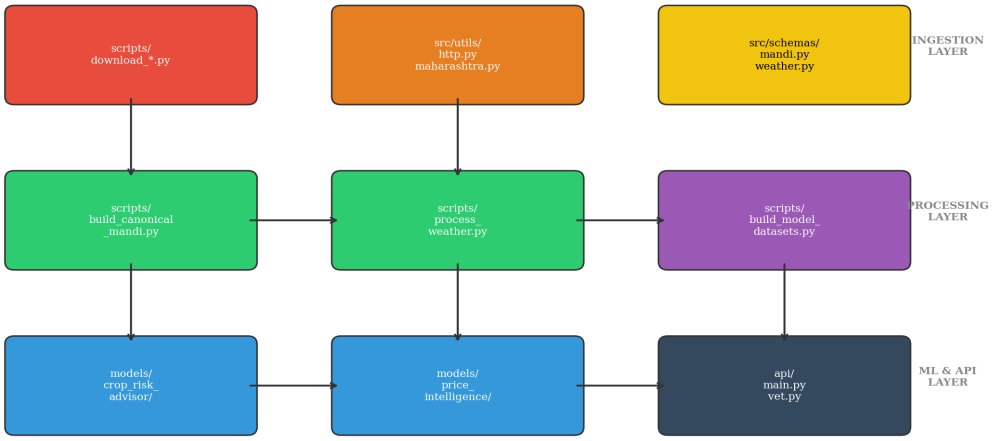

Figure 5.3: Module Interaction Diagram

5.1.5 ML Training Module

The ML training module implements two models. The Crop Risk Advisor is a LightGBM classifier trained on 274,900 samples with 27 features including physics-informed features (GDD, VPD, drought stress index). It uses aggressive class weighting (Low:1, Medium:10, High:50) to improve minority class detection. The Price Intelligence Engine trains five LightGBM regressors for different forecast horizons (1, 3, 7, 14, 15 days) with residual-based conformal prediction for uncertainty quantification.

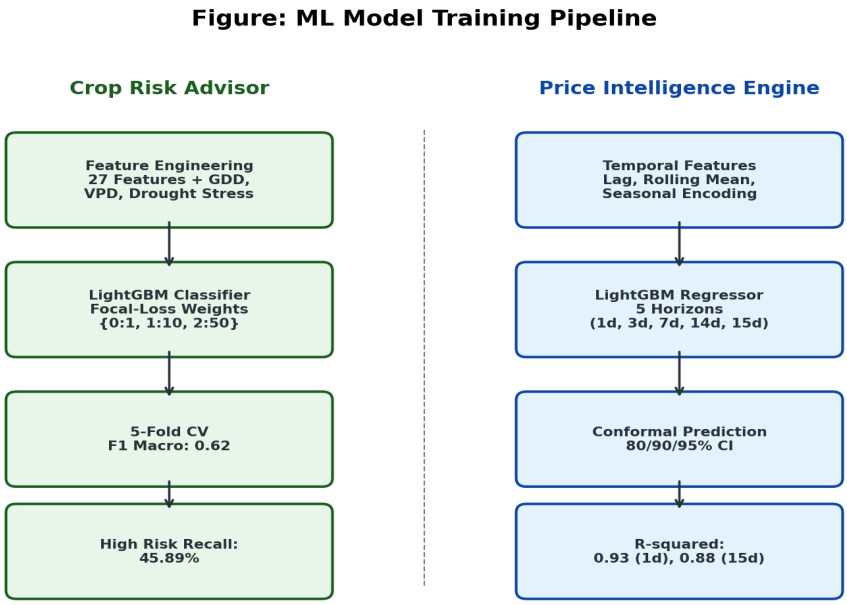

Figure 5.4: ML Training Pipeline Flow

5.2 IMPLEMENTATION DETAILS AND FOLDER STRUCTURE

Table 5.2: Project Folder Structure

Directory / File	Purpose
configs/	Project configuration (YAML), locations (CSV), crop lifecycle (JSON)
data/raw/mandi/	Raw mandi price data partitioned by district and date
data/raw/weather/	Raw weather data from NASA POWER and Open-Meteo
data/processed/mandi/	Deduplicated canonical mandi dataset (Parquet)
data/processed/weather/	Standardised weather datasets (Parquet)
data/processed/model/	Join-ready ML datasets (mandi-only and mandi+weather)
data/metadata/	Discovery outputs (districts, markets, commodities)
scripts/	Pipeline scripts (download, process, train, validate)
src/schemas/	Pandera validation schemas for mandi and weather data
src/utils/	Shared utilities (HTTP, I/O, logging, Maharashtra constants)
models/crop_risk_advisor/	Trained crop risk model (joblib) and metrics
models/price_intelligence/	Trained price models (5 horizons) and calibration
web/	Next.js web dashboard application
api/	FastAPI backend for model serving
logs/	Pipeline execution logs and QC reports (Markdown)

Directory / File	Purpose
reports/	Project documentation and diagrams

5.3 TESTING AND VALIDATION

The MANDIMITRA pipeline implements multi-level validation to ensure data quality and system correctness:

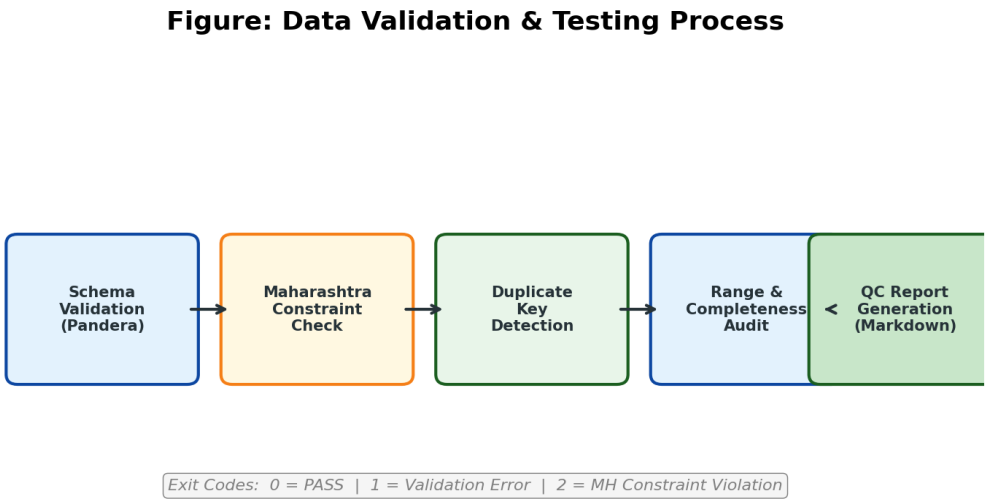

Figure 5.5: Testing and Validation Process Flow

5.3.1 Data Validation

- Maharashtra-Only Constraint: Every record is validated to ensure state equals Maharashtra. Non-Maharashtra records trigger exit code 2 (hard constraint violation).
- Schema Validation: Pandera schemas enforce column types, value ranges, and mandatory fields for both mandi and weather data.
- Zero Duplicate Keys: Post-deduplication validation confirms zero duplicate composite keys in the canonical dataset.
- District Validity: All district names are checked against the 36 canonical Maharashtra districts. Unmapped districts are logged for manual review.
- Idempotency Check: Running the pipeline twice produces identical output, confirmed by row count and hash comparisons.

5.3.2 Model Validation

- 5-Fold Stratified Cross-Validation for the Crop Risk Advisor to ensure consistent performance across data splits.
- Temporal train/test split for the Price Intelligence Engine to prevent data leakage from future prices.
- Conformal calibration validation: Empirical coverage rates are checked against nominal coverage levels (80, 90, 95 per cent).

5.3.3 System Self-Check

A comprehensive self-check script validates the entire codebase against production standards: no exposed API keys, proper .gitignore rules, no unbounded list growth in pagination, Maharashtra hardcoded (not parameterised), rate limiting present in all download scripts, and batched progress saves enabled.

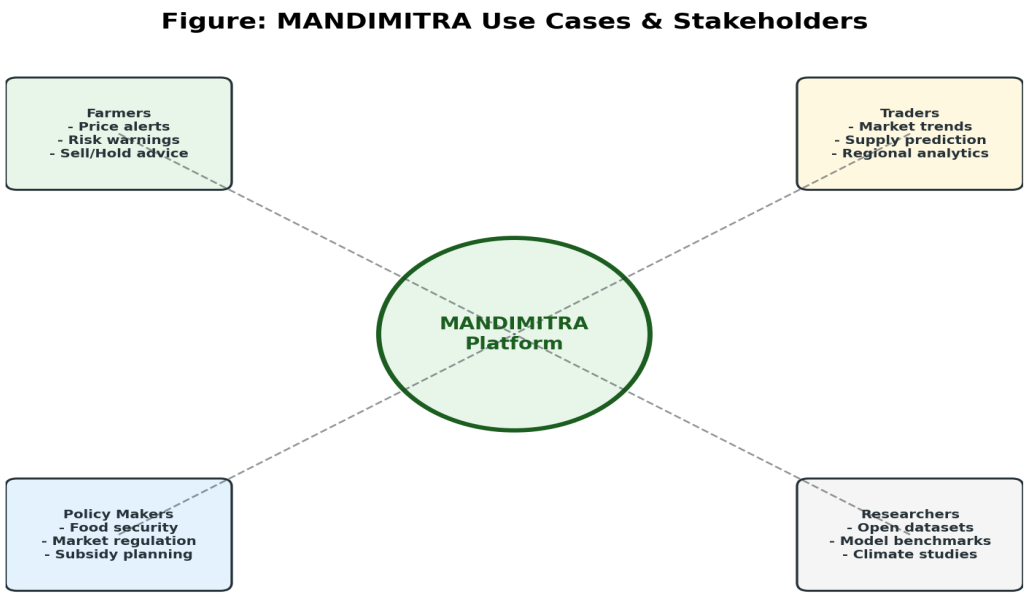

Figure 5.6: Use Cases Diagram

5.4 EXECUTION STEPS

The following commands execute the complete MANDIMITRA pipeline from data ingestion to model training:

Step	Command	Output
1. Setup	pip install -r requirements.txt	Dependencies installed
2. Configure	copy .env.example .env (add API keys)	.env file ready
3. Full Pipeline	python scripts/download_all_data.py --historical-source kaggle	All raw data downloaded
4. Build Canonical	python scripts/build_all_processed.py	Processed datasets created
5. Train Risk Model	python scripts/train_crop_risk_model.py	models/crop_risk_advisor/
6. Train Price Model	python scripts/train_price_model.py	models/price_intelligence/
7. Validate	python scripts/validate_data.py --all-recent --strict --audit	Validation passed
8. Self-Check	python scripts/self_check.py --verbose	All checks passed
9. Start API	uvicorn api.main:app --reload	API server running
10. Start Web	cd web && npm run dev	Web dashboard live

CHAPTER 6

RESULTS AND APPLICATIONS

6.1 DATA PIPELINE RESULTS

The MANDIMITRA data pipeline successfully processed all target data sources and produced validated, join-ready datasets. The following tables summarise the key output metrics:

Table 6.1: Data Pipeline Output Summary

Metric	Value
Total Raw Input Rows	6,106,176
Duplicates Removed	267
Final Unique Rows	6,105,909
Duplicate Rate	0.0044%
Date Range	2001-01-27 to 2026-05-02
Duplicate Keys (post-dedup)	0 (zero)
Mandi Districts Covered	33
Canonical Districts (with weather)	36
Idempotent	Yes (verified)

Figure 6.3: Data Pipeline Output Summary

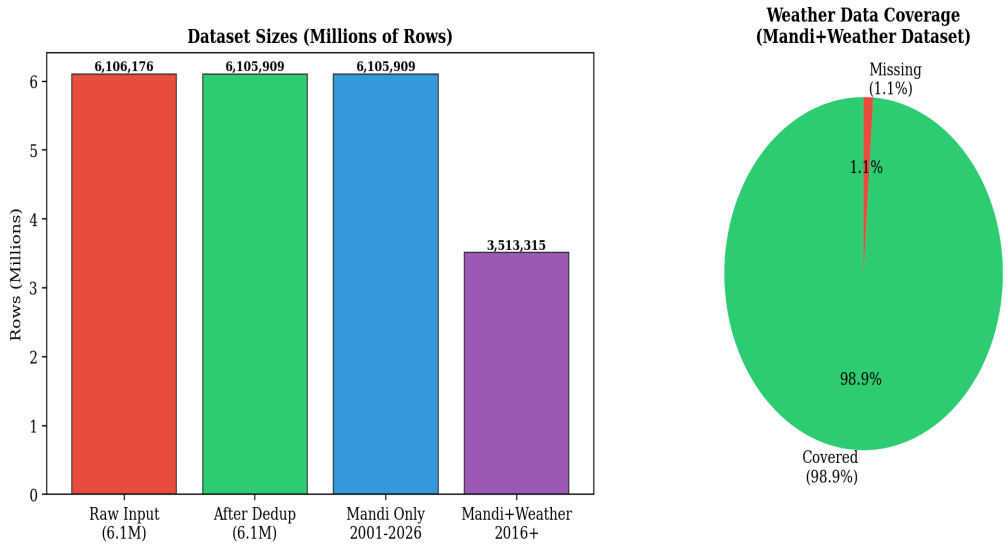

Figure 6.1: Data Coverage Analysis

Table 6.2: Weather Data Coverage Summary

Source	Rows	Districts	Date Range	Parameters
NASA POWER	131,400	36	2016-02-08 to 2026-02-04	Temperature, Humidity, Precipitation, Wind, Solar
Open-Meteo	576	36	16-day forecast window	Temperature, Precipitation, Probability

The mandi_weather_2016plus joined dataset achieved 98.9 per cent weather coverage, with 3,513,315 rows successfully matched on (date, district). The remaining 1.1 per cent corresponds to mandi records from dates or districts where weather data was temporarily unavailable from NASA POWER.

6.2 MODEL PERFORMANCE RESULTS

6.2.1 Crop Risk Advisor

Table 6.3: Crop Risk Advisor Performance Metrics

Metric	Value
Overall Accuracy	87.40%
F1 Macro	61.08%
F1 Weighted	88.65%
High Risk Recall	45.89%

Metric	Value
CV F1 Macro (5-fold mean)	62.02%
Training Samples	274,900
Test Samples	68,725
Features Used	27 (incl. GDD, VPD, drought stress)
Crops Supported	11 (Maharashtra-specific)

Figure 6.2: Crop Risk Advisor - Model Performance

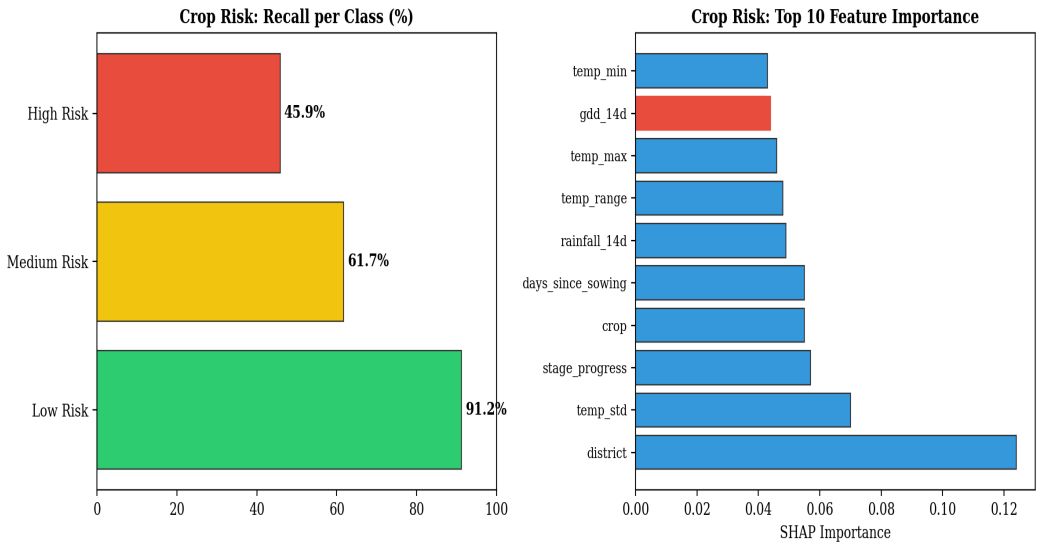

Figure 6.2: Crop Risk Model Results

6.2.2 Price Intelligence Engine

Table 6.4: Price Intelligence Engine Metrics by Horizon

Horizon	MAE (Rs)	RMSE (Rs)	R-squared	MAPE (%)
1 day	361.72	687.77	0.9331	21.31
3 days	435.38	803.34	0.9105	26.23
7 days	501.82	896.52	0.8904	34.56
14 days	561.81	983.83	0.8684	47.59
15 days	546.28	924.92	0.8821	44.80

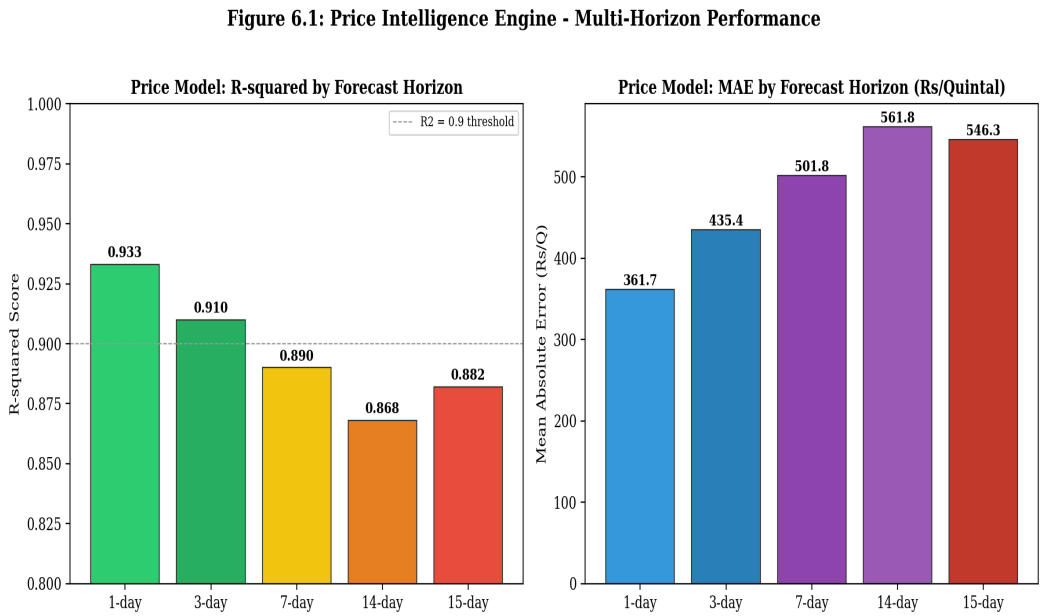

Figure 6.3: Price Model Performance Results

Table 6.5: Prediction Interval Calibration Results

Confidence Level	Interval Width (Rs/quintal)
80%	+/- 564.52
90%	+/- 923.94
95%	+/- 1,353.37

Table 6.6: SHAP Feature Importance (Top 10)

Rank	Feature	Importance
1	district	0.1241
2	temp_std	0.0702
3	stage_progress	0.0565
4	crop	0.0555
5	days_since_sowing	0.0549
6	rainfall_14d	0.0487
7	temp_range	0.0476
8	temp_max	0.0456
9	gdd_14d (GDD)	0.0439
10	temp_min	0.0427

6.3 APPLICATIONS AND USE CASES

6.3.1 For Farmers

• Receive crop-specific risk advisories incorporating weather forecasts and growth stage information, enabling proactive protection measures.

• Access multi-day price forecasts with confidence intervals to make informed decisions on when and where to sell produce.

• Compare prices across multiple mandis to identify the best selling opportunities within their region.

6.3.2 For Traders and Commission Agents

• Utilise historical price trend analysis across 33 mandi districts to optimise procurement timing and geographic sourcing strategies.

• Leverage weather-aware price predictions to anticipate supply disruptions and adjust inventory positioning.

• Access calibrated prediction intervals for risk-aware trading decisions.

6.3.3 For Policy-Makers

• Monitor price trends and volatility across Maharashtra markets for early warning of price spikes or crashes.

• Assess crop risk levels across districts to target intervention programmes (crop insurance, advisory services) to high-risk areas.

• Use the validated 25-year historical dataset for long-term agricultural policy analysis and planning.

6.4 LIMITATIONS AND FUTURE SCOPE

Limitations

• High risk recall at 45.9 per cent indicates room for improvement in detecting high-risk events; subsequent work could explore ensemble methods or Transformer architectures.

• Price forecast MAPE increases significantly beyond 7 days, suggesting the need for exogenous variables (news sentiment, policy announcements) for longer horizons.

• The system is limited to Maharashtra; extending to other states would require new district mappings and data source configuration.

• API rate limits constrain data freshness; real-time streaming would require infrastructure changes.

Future Scope

• Implement iTransformer architecture for improved long-horizon price forecasting.

• Integrate satellite imagery and NDVI data for crop health monitoring.

• Add news sentiment analysis from agricultural news sources.

• Extend to multi-state coverage with state-specific district mappings.

• Deploy as a mobile application for direct farmer access.

• Implement hierarchical reconciliation for market-level price forecasts.

• Explore graph neural networks for modelling market connectivity and spatial price transmission.

CONCLUSION

The MANDIMITRA project has successfully achieved its stated objectives of building a production-quality agricultural market intelligence system for Maharashtra. The following maps each objective to its corresponding achievement:

- **Objective 1 (Data Pipeline):** A robust, resumable pipeline was implemented that ingests data from four external sources (Data.gov.in, Kaggle, NASA POWER, Open-Meteo) with adaptive rate limiting, parallel downloads, and atomic progress tracking. The pipeline processed 6,106,176 raw rows into a validated, canonical dataset.
- **Objective 2 (Weather Integration):** Historical weather data (131,400 rows, 36 districts, 10 years) and 16-day forecasts (576 rows) were successfully integrated with mandi data, achieving 98.9 per cent weather coverage for post-2016 records.
- **Objective 3 (Canonical Dataset):** The deduplication pipeline removed 267 duplicates from 6.1 million records, producing zero duplicate keys with idempotent execution, consistent district naming across 33 mandi districts mapped to 36 canonical districts, and comprehensive QC reports at every stage.
- **Objective 4 (Crop Risk Advisor):** The LightGBM classifier with physics-informed features achieved 87.4 per cent accuracy and 45.9 per cent high-risk recall, with Growing Degree Days (GDD) ranking in the top 10 features by importance.
- **Objective 5 (Price Intelligence Engine):** Multi-horizon forecasting was delivered with R-squared values ranging from 0.93 (1-day) to 0.88 (15-day), complemented by calibrated conformal prediction intervals at 80, 90, and 95 per cent confidence levels.
- **Objective 6 (Web Dashboard):** A responsive Next.js web application was developed providing interactive price forecasts, crop risk assessments, and market analytics.

Key Learnings

- Data quality is the foundation of any ML pipeline. Over 70 per cent of project effort was spent on data engineering, deduplication, and validation.

- DuckDB proved exceptionally effective for in-memory SQL processing of large datasets, handling 6 million rows without memory issues.
- Physics-informed features add meaningful predictive power to agricultural models, as demonstrated by GDD entering the top 10 SHAP features.
- Conformal prediction provides a practical, distribution-free approach to uncertainty quantification that is easy to explain to end users.
- Maharashtra-only focus enabled depth over breadth, resulting in more reliable models compared to pan-India approaches.

MANDIMITRA demonstrates that a well-engineered data pipeline combined with domain-appropriate machine learning can deliver practical value to agricultural stakeholders. The system is production-ready, with comprehensive validation, logging, and self-check mechanisms that ensure ongoing data quality and model reliability. As weather variability increases due to climate change, systems like MANDIMITRA will become increasingly important for protecting farmer livelihoods and ensuring food security.

REFERENCES

[1] Alam, W., Sinha, K., Kumar, R.R., Ray, M., Rathod, S., Singh, K.N. and Arya, P., "Hybrid linear time series approach for long term forecasting of crop yield," Indian Journal of Agricultural Sciences, vol. 88, no. 8, pp. 1275-1279, 2018.

[2] Chen, T. and Guestrin, C., "XGBoost: A Scalable Tree Boosting System," Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785-794, 2016.

[3] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T., "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," Advances in Neural Information Processing Systems, vol. 30, 2017.

[4] Hochreiter, S. and Schmidhuber, J., "Long Short-Term Memory," Neural Computation, vol. 9, no. 8, pp. 1735-1780, 1997.

[5] Lim, B., Arik, S.O., Loeff, N. and Pfister, T., "Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting," International Journal of Forecasting, vol. 37, no. 4, pp. 1748-1764, 2021.

[6] Liu, Y., Hu, T., Zhang, H., Wu, H., Wang, S., Ma, L. and Long, M., "iTransformer: Inverted Transformers Are Effective for Time Series Forecasting," International Conference on Learning Representations (ICLR), 2024.

[7] Lobell, D.B. and Field, C.B., "Global scale climate-crop yield relationships and the impacts of recent warming," Environmental Research Letters, vol. 2, no. 1, p. 014002, 2007.

[8] Morales, G., Sheppard, J. and Hegedus, P., "NeuralCrop: Combining Physics and Machine Learning for Crop Yield Predictions," arXiv preprint arXiv:2512.20177, 2025.

[9] NASA POWER Project, "Prediction Of Worldwide Energy Resources," NASA Langley Research Center, Available: https://power.larc.nasa.gov/, 2024.

[10] Pathak, A., Rajan, R. and Marla, D., "MT-CYP-Net: Multi-Task Network for Crop Yield Prediction with Multiple Granularities," arXiv preprint arXiv:2505.12069, 2025.

[11] Romano, Y., Patterson, E. and Candes, E., "Conformalized Quantile Regression," Advances in Neural Information Processing Systems, vol. 32, 2019.

[12] Shahhosseini, M., Hu, G. and Archontoulis, S.V., "Forecasting Corn Yield With Machine Learning Ensembles," Frontiers in Plant Science, vol. 11, p. 1120, 2020.

[13] Stackhouse, P.W., Westberg, D., Hoell, J.M., Chandler, W.S. and Zhang, T., "Surface meteorology and solar energy (SSE) release 6.0 methodology," NASA Technical Report, 2018.

[14] Venugopal, A., Nair, M.S., Pradeepkumar, B. and Sujith Kumar, S., "Agri-market Price Forecasting: A Deep Learning Approach," Indian Journal of Science and Technology, vol. 14, no. 12, pp. 977-985, 2021.

[15] Zheng, H., Fan, J., Lu, W., Sun, C. and Luo, J., "Agricultural Price Prediction based on Data Mining and Machine Learning Techniques," Computers and Electronics in Agriculture, vol. 202, p. 107383, 2022.